Question	Answer	Marks	Partial Marks	Guidance
22	$2x^2 - 5x - 15$ [= 0] or $y^2 - 34y + 144$ [= 0] oe	**M2**	**M1** for $4x + 12 = 2x^2 - x - 3$ or better or $y = 2\left(\frac{y-12}{4}\right)^2 - \left(\frac{y-12}{4}\right) - 3$ or better	For M2 accept equivalent 3 term expression
	$\frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-15)}}{2(2)}$ oe or for $\frac{5}{4} \pm \sqrt{\frac{15}{2} + \left(\frac{5}{4}\right)^2}$ oe	**M2**	**FT** *their* 3-term quadratic in x or y or **M1** for $\sqrt{(-5)^2 - 4(2)(-15)}$ or better or for $\frac{-(-5) + \sqrt{q}}{2 \times 2}$ oe or $\frac{-(-5) - \sqrt{q}}{2 \times 2}$ oe or for $\left(x - \frac{5}{4}\right)^2$	Do not FT from $2x^2 - x - 3$ [= 0] For M2, do not accept $\frac{5 \pm \sqrt{145}}{4}$ Sq root to cover – sign, division to cover ± sign, but can recover with working seen
	(4.26 , 29.04) and (– 1.76, 4.96)	**B2**	**B1** for one correct pair or both x-values correct or both y – values correct	Condone more accurate values for B2 and B1 (4.260…, 29.041 to 29.042) and (– 1.760… , 4.958…) Correct or more accurate values may be seen in working for B2 or B1 If $x = 4.26$ and $x = -1.76$ seen in working and then transcribed incorrectly to answer line as y values then allow B1 Similarly if working with y values first